SC4020 Project 2
Technical Review Report

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Task 1

Analysis of Symptom Co-occurrence Patterns

The objective of this task is to perform association rule mining using the Apriori algorithm to identify frequently co-occurring symptoms combinations across the entire collection of disease profiles. The dataset is restructured into a standard transaction format, where each unique disease profile (N=41) represents a single transaction, and each unique symptom represents an item in the basket. Since the Apriori algorithm requires binary input, the analysis is strictly focused on the presence or absence of a symptom within a disease profile, meaning whether or not a symptom is listed multiple times for a disease, it is collapsed into a single instance of presence.

1.1 Data preparation and cleaning

Data Cleaning is performed by removing null values, converting strings to lowercase, and removing underscores. The raw data is then melted from its wide format (Symptom_1 to Symptom_17) into a long format of (Disease, Symptom).

1.2 Symptom Normalisation (Synonym Mapping)

The reason for normalisation is because the dataset uses heterogeneous terminology, where 'high fever', 'mild fever', 'pyrexia' are distinct entries but represent the same underlying clinical concept: 'fever'. Without normalisation, the Apriori algorithm would treat these as separate, rare items, failing to recognize the pattern. A custom **synonym_map** is applied to cluster the semantically equivalent terms. This step reduces the total number of unique symptoms from 131 to 80 and helps to consolidate the rare but same variations, improving the clustering potential for the Apriori algorithm.

1.3 One-hot Encoding

The cleaned and normalised long-format data as mentioned under '1.1 Data preparation and cleaning' is one-hot encoded to create the binary matrix. The matrix is grouped by 'Disease' and the .max() operation is applied. This ensures that any symptom present in a disease is marked as '1' regardless of how many symptom columns it occupies.

1.4 Apriori algorithm and initial results

We set the minimum support threshold to be 0.15 (15%), where a pattern is considered 'frequent' only if it appears in at least 15% of all disease profiles. Based on the original N=41 diseases, min_sup = 0.15 results in a minimum absolute count of 7 diseases ($\lceil 41 \times 0.15 \rceil = 6.15$, rounded up to 7).

```
--- Top Frequent Co-occurring Symptom Combinations ---
These are symptom combinations that frequently appear together in the same disease profile
   support                              itemsets
0   0.2195             (vomiting, abdominal pain)
1   0.2195                      (fatigue, fever)
2   0.1951                   (fatigue, vomiting)
3   0.1951           (loss of appetite, vomiting)
4   0.1707       (yellowish skin, abdominal pain)
5   0.1707            (fatigue, loss of appetite)
6   0.1707                      (vomiting, fever)
7   0.1707  (loss of appetite, yellowing of eyes)
8   0.1707             (yellowish skin, vomiting)
```

1.5 Data augmentation for robustness

Given the results above, only 8 frequent itemsets were found with support >= 15%. With only 41 samples, the support calculation is highly susceptible to noise or accidental co-occurence.

- Low margin of error: A pattern appearing in 7 diseases is deemed frequent. If one of those 7 instances is removed, perhaps due to misentry of data or atypical patient profile), then the pattern's support falls below the 15% threshold. This sensitivity indicates low confidence.
- Furthermore, since the total transaction count (N=41) is small, even a small variation in the sample will lead to large fluctuations in relative support.

Hence we will be stochastically augmenting the dataset by mirroring the dataset, and then injecting noise where a small, random amount of data (5%) is flipped (symptoms are randomly added or dropped). This would allow us to both increase the number of transactions, while also simulating real-world noise such as slight biological variability in patients.

Result after augmentation (The absolute count is now 13 (82 * 0.15 = 12.3), and total transactions is 82 (N=41 Original + N=41 Noisy Mirror):

```
--- Top 10 Frequent Co-occurring Symptom Combinations ---
These are symptom combinations that frequently appear together in the same disease profile.
   support                        itemsets
0   0.2073            (vomiting, abdominal pain)
1   0.2073                     (fatigue, fever)
2   0.1829          (loss of appetite, vomiting)
3   0.1707       (yellowish skin, abdominal pain)
4   0.1707                  (fatigue, vomiting)
5   0.1585                    (vomiting, fever)
6   0.1585  (loss of appetite, yellowing of eyes)
7   0.1585            (yellowish skin, vomiting)
```

Analysis of robustness:
The fact that the top 7 out of the initial top datasets remained as frequent co-occurring symptoms, despite having 5% of their supportive transactions randomly disrupted proves the degree of robustness. Although the support for {fatigue, fever} has dropped from 0.2195 to 0.2073, their survival above the min_sup = 0.15 threshold demonstrates that these patterns are not fragile, even when subjected to significant simulated noise and higher absolute frequency requirement.

Hence, augmenting data is essential in this case, because the low initial transaction count (N=41) inherently leads to lower statistical confidence in any of the discovered combinations, regardless of relative support percentage (min_sup). Augmenting the data with noisy mirrored dataset addresses this problem by significantly increasing the statistical denominator while introducing real-world variability. It is good to prove that the discovered frequent co-occurring symptoms are truly common signals that persist, transforming them from fragile observations, to more robust findings.

Task 2

Mining Cancer Feature Patterns

The objective of this task is to analyse sequential patterns and relationships between features in breast cancer data, to identify key characteristics that differentiate malignant cases from benign ones.

2.1 Data preprocessing

Irrelevant columns for analysis are first removed, specifically the columns [“id”] and [“Unnamed: 32”]. The target column [“diagnosis”] is mapped from ‘M’ and ‘B’ to 1 and 0 respectively.

The numerical features are then discretized into categorical sequences using KBinsDiscretizer with multiple strategies (uniform, quantile, k-means). 3 bins were used, to represent “low”, “medium” and “high”.

The sequence semantics used to rank features per patient is **mutual information (MI) with respect to diagnosis**, to establish a fixed, global ranking of features based on their importance in predicting the diagnosis.

We decided to use MI to rank features instead of z-score, because it directly measures how much information a feature provides about the target variable (diagnosis). Furthermore, MI is able to capture linear and non-linear relationships between features and the target variable. Z score on the other hand is more useful for standardizing the features and does not directly provide information about a feature’s ability in predicting the diagnosis.

The **top-10 features** are then grouped as ordered itemsets with max sequence length L=10 and max-gap=1.

```
Top 10 Features by Mutual Information:
perimeter_worst         0.471842
area_worst              0.464313
radius_worst            0.451230
concave points_mean     0.438806
concave points_worst    0.436255
perimeter_mean          0.402361
concavity_mean          0.375447
radius_mean             0.362276
area_mean               0.360023
area_se                 0.340759
```

Sequence construction is then performed for each patient by taking the discretized values of their top 10 most important features, ordered by the MI ranking as shown above. For each feature, we retrieve its discretized value (e.g., "high", "medium") and create a sequence item combining the feature name and its discretized value. For example: ['perimeter_worst_high', 'radius_worst_medium', 'concave_points_worst_high']. This sequence represents how different cancer features manifest in that particular patient, with the most informative features appearing first.

2.2 Data Analysis

To facilitate comparison and pattern discovery, we separate the sequences into two categories based on the diagnosis:

- Malignant sequences: Sequences associated with cancer (diagnosis = 1).
- Benign sequences: Sequences associated with no cancer (diagnosis = 0).

We implemented the Generalized Sequential Pattern (GSP) algorithm for sequential pattern mining. After mining the sequences, we performed data postprocessing to only get the closed frequent sequences, to reduce redundancy. The tables below show the closed frequent sequential patterns based on support count, for malignant and benign tumours respectively.

To first show and interpret the initial results from pattern mining, we used quantile binning as an example. Other strategies will be explored below.

A closed frequent itemset is a frequent itemset for which no superset has the same support. We chose to focus on closed frequent itemsets because they provide a concise, non-redundant summary of all frequent patterns in the dataset while retaining the support information of all included subsets. This approach reduces redundancy, highlights informative co-occurrences, and ensures that all relevant frequency information is preserved for interpretation.

Rank	**Pattern Sequence**	**Support**
1	radius_worst_high	0.84
2	concave points_worst_high	0.84
3	perimeter_worst_high	0.84
4	area_worst_high → radius_worst_high	0.83
5	perimeter_worst_high → radius_worst_high	0.82
6	concave points_mean_high	0.82
7	perimeter_worst_high → area_worst_high → radius_worst_high	0.81
8	perimeter_mean_high	0.80
9	area_mean_high	0.79
10	concavity_mean_high	0.79

Table 1: Top 10 Closed Frequent Feature Patterns in Malignant Tumours
(Parameters: Quantile Binning, Min Support=30%)

For malignant tumours, the top patterns are dominated by high values of key features such a**s radius_worst, concave points_worst, perimeter_worst, area_worst, and concavity_mean.**

Many patterns are **single features** with very high support (0.79–0.84), indicating these features individually occur in the majority of malignant sequences.

The top patterns mostly involve **high values** of features like:

- radius_worst_high → the largest tumor radius measured.
- concave points_worst_high → degree of irregularity in the tumor boundary.
- perimeter_worst_high and area_worst_high → overall size of the tumor.
- concavity_mean_high → average concavity, indicating irregular shapes.

These features being high suggests that malignant tumours tend to be ***larger, irregular, and more concave than benign tumours***. Multi-feature patterns (like **area_worst_high → radius_worst_high**) show that size and irregularity co-occur, reflecting more aggressive tumor growth. High support values (0.79–0.84) indicate these characteristics are very common across malignant cases.

Rank	**Pattern Sequence**	**Support**

1	area_worst_low	0.53
2	concave points_mean_low	0.53
3	perimeter_worst_low	0.53
4	concave points_worst_low	0.52
5	radius_worst_low	0.52
6	perimeter_mean_low	0.52
7	area_mean_low	0.52
8	concavity_mean_low	0.52
9	area_worst_low → radius_worst_low	0.52
10	radius_mean_low → area_mean_low	0.51

Table 2: Top 10 Closed Frequent Feature Patterns in Benign Tumours
(Parameters: Quantile Binning, Min Support=30%)

For benign tumours, the top patterns mostly involve **low values** of features such as:

- area_worst_low, radius_worst_low → smaller size.
- concave points_mean_low, concavity_mean_low → more regular, smoother tumor shape.
- Multi-feature patterns (like area_worst_low → radius_worst_low) show **co-occurrence of small size and regular shape**.

Benign tumours tend to be ***smaller and more regular*** in shape, reflecting their non-aggressive nature. Support values are slightly lower (0.51–0.53), indicating more variability among benign cases, which is biologically consistent since benign tumors can vary in size without being malignant.

Overall, malignant tumors are characterized by extreme high values in size and shape irregularity, while benign tumors show extreme low values.

Sensitivity check across binning strategies

Exact maximal patterns change across binning methods due to discretization. However across binning strategies, core features are robust, even if the full sequence differs.

Comparison of Top 5 Closed Frequent Patterns in Malignant Tumours Across Binning Methods

Rank	Quantile Binning	Support	Uniform Binning	Support	KMeans Binning	Support
1	**radius_worst_high**	0.84	area_se_low	0.98	concave points_mean_medium	0.69
2	concave points_worst_high	0.84	perimeter_mean_medium	0.75	concavity_mean_medium	0.62
3	perimeter_worst_high	0.84	perimeter_mean_medium → area_se_low	0.75	concave points_worst_high	0.61
4	area_worst_high → radius_worst_high	0.83	radius_mean_medium	0.75	area_mean_medium	0.58
5	perimeter_worst_high → radius_worst_high	0.82	radius_mean_medium → area_se_low	0.74	area_worst_medium	0.58

Table 3: Comparison across binning methods for malignant tumours

Comparison of Top 5 Closed Frequent Patterns in Benign Tumours Across Binning Methods

Rank	Quantile Binning	Support	Uniform Binning	Support	KMeans Binning	Support
1	**area_worst_low**	0.53	**area_worst_low** → area_se_low	1.00	area_se_low	1.00
2	concave points_mean_low	0.53	**area_worst_low** → area_mean_low → area_se_low	1.00	**area_worst_low**	1.00
3	perimeter_worst_low	0.53	perimeter_worst_low → **area_worst_low** → area_mean_low → area_se_low	0.99	**area_worst_low** → area_se_low	0.99
4	concave points_worst_low	0.52	**area_worst_low** → concave points_mean_low → area_se_low	0.98	**area_worst_low** → area_mean_low	0.98
5	radius_worst_low	0.52	perimeter_worst_low → **area_worst_low** → radius_worst_low → area_mean_low → area_se_low	0.98	**area_worst_low** → area_mean_low → area_se_low	0.98

Table 4: Comparison across binning methods for benign tumours

Binning Method	**Primary Feature Focus**	**Key Interpretation**
Quantile Binning	(radius_worst_high, perimeter_worst_high)	By creating bins with equal observation counts, this method emphasizes the most **extreme percentiles**. The resulting patterns are long, highly supported, and clinically intuitive: **malignancy is defined by extreme, worst-case morphological metrics**. This provides the strongest evidence for feature progression
Uniform Binning	(area_se_low, perimeter_mean_medium)	By creating bins of equal width across the feature range, this method captures less of the extreme high-end signal. The high support for area_se_low (0.98) is counter-intuitive for a malignant tumor (which should have high standard error, indicating irregularity). This suggests that **uniform binning fails to adequately capture the discriminatory power of the malignant features**.
KMeans Binning	(concave points_mean_medium, concavity_mean_medium)	KMeans identifies patterns that represent the **most typical feature cluster centroids.** While the top patterns are medium, the third-ranked feature is **concave points_worst_high** (0.61), showing that the extreme irregularity is still strongly represented in one of the natural feature clusters.

This suggests that the results are sensitive to the binning method of choice due to the way each method partitions the continuous feature space, critically affecting the resulting patterns' support and clinical relevance.

Task 3

3.1 Problem Definition

When the coronavirus hit, healthcare systems worldwide faced tremendous pressures from surging patient volumes, limited medical staff and the need for timely decision making. With travel and globalisation becoming more prominent, there is no doubt of future cross border disease transmissions overloading hospitals again (Terefe et al., 2023).

A key challenge lies in triaging patients effectively, determining the urgency of their diseases. Delays in making such decisions can be fatal as high risk patients may miss timely intervention while low risk patients may face unnecessary tests resulting in inefficient allocation of hospital resources. This surfaces an urgent need for applications that can support healthcare professionals by prioritizing patients based on both disease risk and disease urgency.

We thus propose a multi facet healthcare application for disease prediction and risk assessment targeting diseases ranging from chronic conditions like cancer to transmissible viruses.

3.2 Objective

Our goal is to develop a general healthcare application with 2 entry points:

A) Symptom checker: predicts likely diseases based on a patient's symptoms
B) Cancer Pathway: predicts if a cancer patient has malignant or benign cancer

This allows hospitals to classify patients into low to high urgency and maximise their allocation of resources.

We created a multimodal system using neural networks for cancer prediction and random forest classifier for the symptom checker.

3.3 Module 1: Breast Cancer Urgency Predictor

3.3.1 Model Architecture

This module is a Multi Layer Perceptron which takes in 30 continuous features to predict the probability of malignant cancer.

3.3.2 Data Pipeline

Preprocessing: The dataset of 30 features were divided into train, test and validation using a 60-20-20 split respectively. The diagnoses were one-hot encoded with 0 representing benign and 1 representing malignant cancer. All 30 features were standardized using StandardScaler() which was fitted on the train split.

3.3.3 Model Training and Evaluation

Training and model parameters: The model was trained over 30 epochs while monitoring validation performance to avoid overfitting as data is limited.

Component	**Setting**	**Explanation**
Loss function	nn.BCEWithLogitsLoss()	Standard choice for binary classification in PyTorch.
Optimizer	Adam	Robust and widely used optimizer that adapts learning rate
Learning Rate	0.001	Standard initial learning rate for Adam
Weight Decay	1e - 4	L2 regularization is applied to the weights to discourage large parameter values as overfitting is likely with smaller datasets
Scheduler	ReduceLROnPlateau Patience: 3 Factor: 0.5	Monitors validation loss to escape local minima and fine tune parameters after initial rapid progress.

Model Performance

Performance on Test Split:

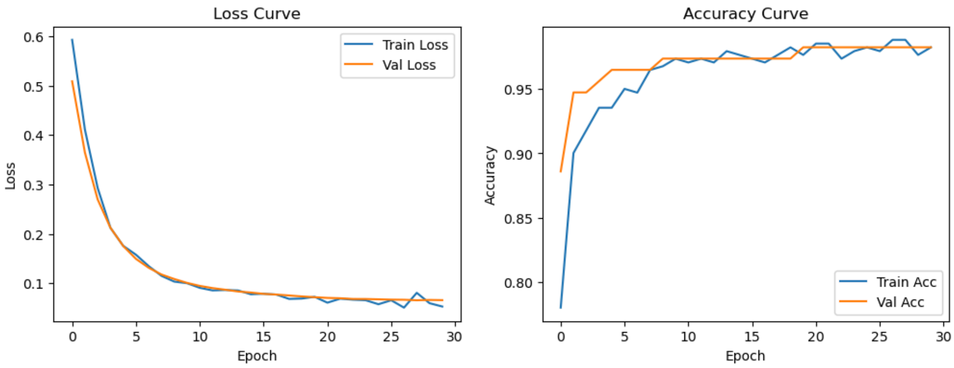
```
Classification Report:
              precision    recall  f1-score   support

           0       0.97      1.00      0.99        72
           1       1.00      0.95      0.98        42

    accuracy                           0.98       114
   macro avg       0.99      0.98      0.98       114
weighted avg       0.98      0.98      0.98       114

Test Accuracy: 0.9824561403508771
```

Train vs Validation Loss and Accuracy Curves



3.4 Module 2: Symptom Checker for Triaging

Goal: Given a patient's self-reported symptoms, predict the most likely disease(s) and map them to an urgency tier to support front-desk triage.

Data and features: We reuse the cleaned, normalised symptom vocabulary from Task 1. Each disease profile (N = 41) is represented as a binary one-hot vector over these 131 symptoms (presence/absence). For modelling we perform a stratified 80/20 split with a fixed random seed and report both Top-1 and Top-3 accuracy.

3.4.1 Model training

We benchmarked Logistic Regression and a Random Forest (RF). The final classifier that we used is **Random Forest Classifier** (n_estimators=200, bootstrap=True). RF was chosen because it handles sparse, non-linear symptom interactions without feature scaling, provides calibrated-ish probability scores via averaging trees, and remains robust to small, spurious symptoms.

Training and evaluation: We benchmarked Logistic Regression against a Random Forest and selected a RandomForestClassifier with 200 trees, class_weight set to "balanced," and bootstrap enabled. This choice handles sparse, non-linear interactions without feature scaling, produces reasonably calibrated probability estimates through tree averaging, and is robust to a few spurious features. Training uses standard Gini impurity minimisation.

Result: On the held-out test set the model attains **100% Top-1 accuracy** (hence Top-3=100%). Precision/recall/F1 are 1.00 for all 41 classes; the confusion matrix is perfectly diagonal (Fig. 3.4a). A **5-fold learning curve** shows both training and cross-validated accuracy reach 1.00 with ~1.2k training examples and stay flat thereafter, indicating trivial separability (Fig. 3.4b).

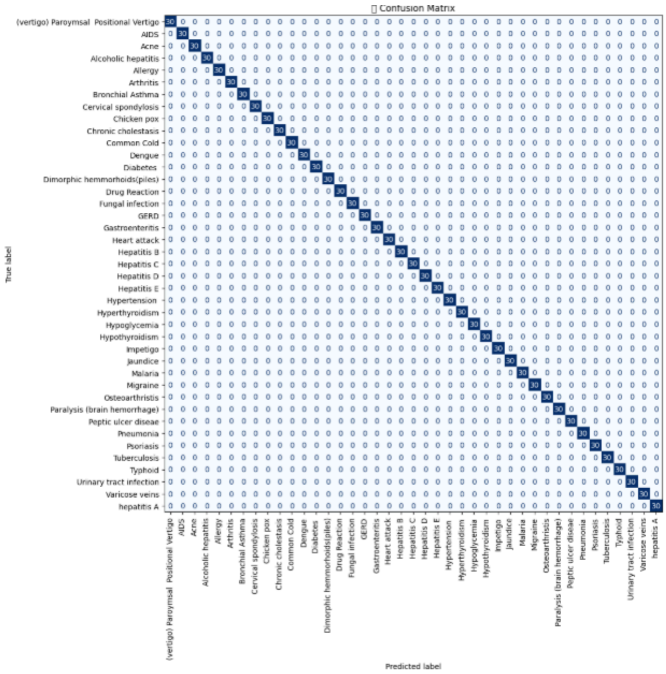


Fig 3.4a. Confusion matrix

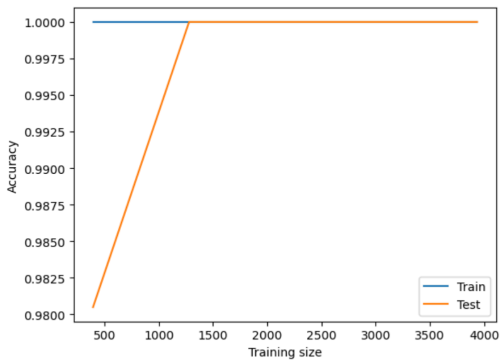


Fig 3.4b 5-fold learning curve

3.4.2 Evaluation

Inference Pipeline: At inference, free-text symptoms are normalized (lower-cased, punctuation-stripped, and synonym-mapped) and vectorized into a 131-dimensional one-hot vector. The Random Forest outputs class probabilities, presenting the Top-3 predicted diseases.

We combine the top prediction with a symptom severity score, computed as the dot product of the one-hot vector with per-symptom weights. We use tuned cut-offs (18.65 and 26.05) to define tiers: Low if the total score < 18.65 and the top disease is not in a predefined urgent list; Moderate if 18.65–26.04 without urgent disease; and High if ≥ 26.05 or the top disease is in the urgent list.

Robustness Notes. To probe brittleness, we ran a small noise test where 1–2 random symptoms were flipped per case (≈5% perturbation). Top-3 suggestions were stable; Top-1 changed only when distinctive, disease-defining symptoms were removed—expected behaviour for binary, rule-like profiles.

Conclusion and key insights: The Symptom Classifier achieved 100% accuracy on both train and test (with perfect precision, recall, and F1 across all 41 classes). It exposes a dataset limitation. The dataset is synthetic or over-structured, each disease has a near-unique symptom signature with little overlap. There is no realistic variation (negations, severities, partial presentations), so the model effectively memorises look-up patterns in the one-hot space. Consequently, many simple models like Random Forest reach perfect scores.

3.5 Results

You may refer to this Streamlit application for a demonstration of our healthcare triaging application:
https://healthcare-triaging-application.streamlit.app/

References

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- https://www.kaggle.com/code/ctrlsari/symptom-classifier-w-live-demo-random-forest
- http://rasbt.github.io/mlxtend/user_guide/frequent_patterns/apriori/